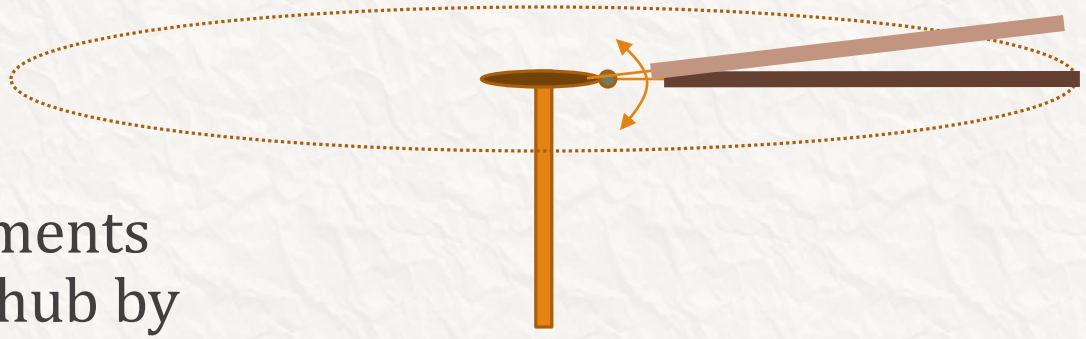
Rotary Wing Aerodynamics (AE 667)

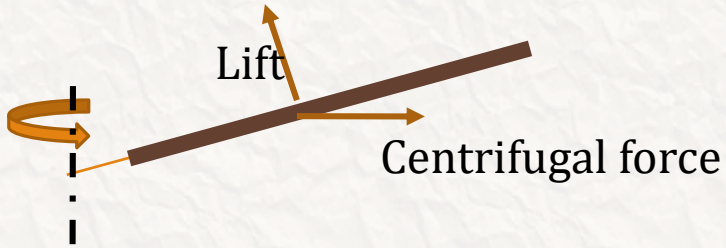


Coning angle

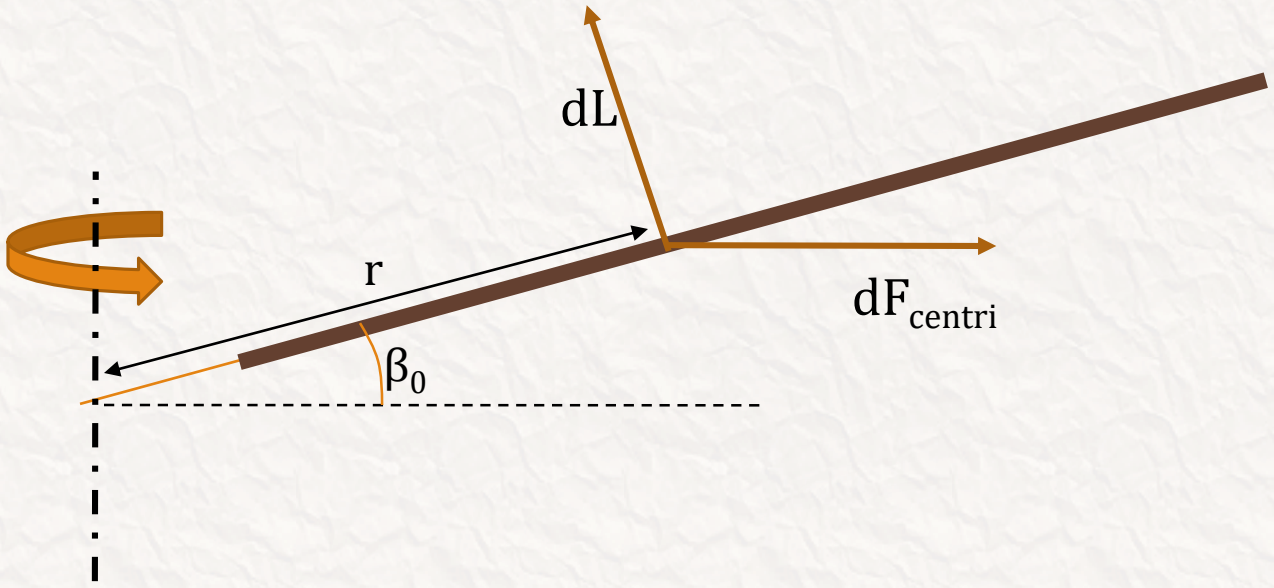
Flapping Hinge



- Flapping hinge
  used to keep bending moments low at the blade root and hub by allowing blades to flap
- Aerodynamic lift on blade causes upward flapping motion
- Centrifugal forces cause downward flapping motion



Flapping Hinge



- Here, the hinge is assumed to be at the center
- Generally there is hinge offset, implications on control

Flapping Hinge (Aerodynamic Moment)

- From BE theory,

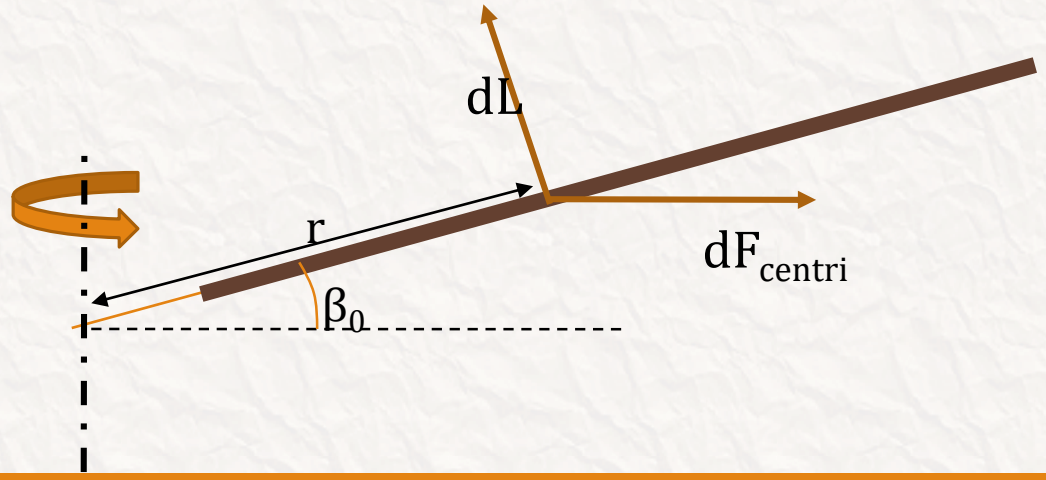
$$dL \approx \frac{1}{2}\rho c(\Omega r)^2 C_l dr$$

- Moment:

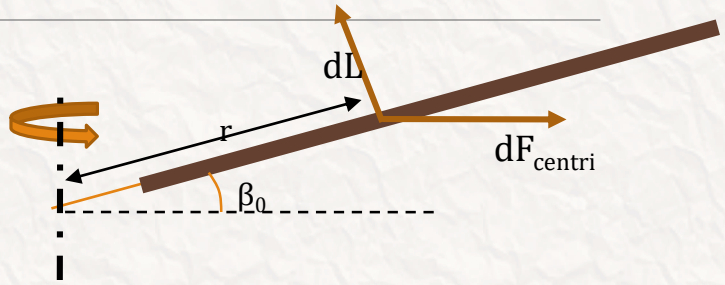
$$r.dL \approx \frac{1}{2}\rho c(\Omega r)^2 r C_l dr$$

- Integrating moment:

$$\int_{r=0}^{r=R} \frac{1}{2}\rho c(\Omega r)^2 r C_l dr$$



Flapping Hinge (Centrifugal Moment)



- $dF_{centri} = \frac{(\Omega r \cos(\beta_0))^2 dm}{r} \approx \Omega^2 r dm$

- $Moment = r \sin(\beta_0) dF_{centri} \approx \Omega^2 r^2 \beta_0 dm$

$$\text{Integrating moment}: \int_{r=0}^{r=R} \Omega^2 r^2 \beta_0 dm \equiv I\Omega^2\beta_0$$

- Here, dm is the mass of the strip dr
  & I is mass moment of inertia of the rotor blade

Equilibrium Flap Angle β_0

At equilibrium, moment due to aerodynamic lift
= moment due to centrifugal forces

$$I\Omega^2\beta_0 = \int_{r=0}^{r=R} \frac{1}{2}\rho c(\Omega r)^2 r C_l dr$$

$$\beta_0 = \frac{\int_{r=0}^{r=R} \frac{1}{2}\rho c r^3 C_l dr}{I} = \frac{\rho a c R^4}{I} \int_{r=0}^{r=R} \left(\frac{r}{R}\right)^3 \alpha_{effective} d\left(\frac{r}{R}\right)$$

Lock Number (Υ)

Lock Number (γ)

- Measure of aerodynamic moment to centrifugal moment on rotor blade
- γ : 8 – 10 for articulated rotors
  γ : 5 – 6 for hingeless rotors
- Has implications on auto-rotation